

AMATH 403 HW 5

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Question 1: 4.3.18. Use separation of variables to solve the Helmholtz boundary value problem $\Delta u = u$, $u(x, 0) = 0$, $u(x, 1) = f(x)$, $u(0, y) = 0$, $u(1, y) = 0$, on the unit square $0 < x, y < 1$.

Using separation of variables:

$$u(x, y) = v(x)w(y)$$

Helmholtz equation:

$$\Delta u = u$$

$$u_{xx} + u_{yy} = u$$

Using substitution:

$$u_{xx} = v''(x)w(y)$$

$$u_{yy} = v(x)w''(y)$$

$$v''(x)w(y) + v(x)w''(y) = v(x)w(y)$$

Divide by $v(x)w(y)$

$$\frac{v''(x)}{v(x)} + \frac{w''(y)}{w(y)} = 1$$

$$\frac{v''(x)}{v(x)} = 1 - \frac{w''(y)}{w(y)} = \lambda$$

Finding the first ODE:

$$v''(x) = \lambda v(x)$$

$$v''(x) - \lambda v(x) = 0$$

Finding the second ODE:

$$\frac{-w''(y)}{w(y)} = \lambda - 1$$

$$-w''(y) = (\lambda - 1)w(y)$$

$$w''(y) - (1 - \lambda)w(y) = 0$$

Based on the given boundary conditions

$$u(0, y), v(0)w(y) = 0 \implies v(0) = 0$$

$$u(1, y), v(1)w(y) = 0 \implies v(1) = 0$$

Solving $v''(x) - v(x) = 0$

Case $\lambda = 0$:

$$\begin{aligned}v''(x) &= 0 \\v(x) &= Ax + B \\v(0) &= B = 0 \\v(1) &= A(1) + 0 = 0 \\v(x) &= 0\end{aligned}$$

Case 2 $\lambda > 0, \lambda = k^2$:

$$\begin{aligned}v''(x) - k^2v(x) &= 0 \\v(x) &= Ae^{kx} + Be^{-kx} \\v(0) &= Ae^0 + Be^0 = 0 \\v(1) &= Ae^k + Be^{-k} = 0\end{aligned}$$

Case $\lambda < 0, \lambda = -k^2$

$$\begin{aligned}v''(x) + k^2v(x) &= 0 \\v(x) &= A \cos(kx) + B \sin(kx) \\v(0) &= A(1) + B(0) = 0 \implies A = 0 \\v(1) &= (0) \cos(k) + B \sin(k) = 0 \\ \sin(k) &= 0 \implies k = n\pi, \quad n = 1, 2, 3, \dots\end{aligned}$$

Solving $w(y)$:

$$\begin{aligned}\lambda_n &= -(n\pi)^2 \\w''(y) - (1 - \lambda_n)w(y) &= 0 \\w''(y) - (1 + n^2\pi^2)w(y) &= 0 \\\mu_n &= \sqrt{1 + n^2\pi^2} \\w_n(y) &= C_n \cosh(\mu_n y) + D_n \sinh(\mu_n y) \\u(x, 0) &= 0 \\w(0) &= C_n \cosh(0) + D_n \sinh(0) = 0 \\C_n(1) + D_n(0) &\implies C_n = 0 \\W_n(y) &= D_n \sinh(y\sqrt{1 + n^2\pi^2})\end{aligned}$$

By superposition:

$$u(x, y) = \sum_{n=1}^{\infty} b_n \sin(n) \sinh(y\sqrt{1 + n^2\pi^2})$$

Finding b_n :

$$u(x, 1) = f(x)$$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(n) \sinh(y\sqrt{1+n^2\pi^2})$$

$$b_n \sinh(y\sqrt{1+n^2\pi^2}) = 2 \int_0^1 f(x) \sin(nx\pi) dx$$

$$b_n = \frac{2}{\sinh(y\sqrt{1+n^2\pi^2})} \int_0^1 f(x) \sin(nx\pi) dx$$

Question 2: Give an example of functions f and g , along with an explicit solution u , such that u attains its maximum value at an interior point of the domain D .

$$-\Delta u = f$$

In the unit open disk

$$D := \{(x, y) : x^2 + y^2 < 1\}$$

Dirichlet boundary on the unit circle C which is the boundary

$$u = g, \quad \partial D = C = \{(x, y) : x^2 + y^2 = 1\}$$

Use a function that has its peak at the interior:

$$1 - x^2 - y^2$$

Evaluate at boundary:

$$u(x, y) = 1 - (x^2 + y^2)$$

$$= 1 - 1$$

$$= 0$$

$$g(x, y) = 0$$

$$\Delta u = f$$

$$-(u_{xx} + u_{yy}) = f$$

$$-(-2 + -2) = f$$

$$f(x, y) = 4$$

$$u(x, y) = 1 - x^2 - y^2$$

$$f(x, y) = 4$$

$$g(x, y) = 0$$

Question 3: Textbook Problem 7.1.5

(a) Prove $\hat{\delta}(k) = \frac{1}{\sqrt{2\pi}}$

$$\hat{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$$

$$\hat{\delta}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(x) e^{-ikx} dx$$

$$f(x) = e^{-ikx}$$
$$\int_{-\infty}^{\infty} \delta(x) f(x) dx$$

Using the sifting property:

$$\int_{-\infty}^{\infty} \delta(x) f(x) dx = f(0)$$

$$\int_{-\infty}^{\infty} \delta(x) e^{-ikx} dx = f(0)$$

$$f(0) = e^{-ik0} = e^0 = 1$$

Thus

$$\delta(\hat{k}) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(x) f(x) dx = \frac{1}{\sqrt{2\pi}}$$

$$\delta(\hat{k}) = \frac{1}{\sqrt{2\pi}}(1) = \frac{1}{\sqrt{2\pi}}$$

(b) Use the Fourier inversion formula and part (a) to find the Fourier transform of $e^{i\omega x}$, $\omega \in \mathbb{R}$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(k) e^{ikx} dk$$

We know $f(x) = \delta(x)$, $\hat{f}(k) = \frac{1}{\sqrt{2\pi}}$

$$\delta(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{ikx} dk$$

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dk$$

Change Variables

$$\delta(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iku} dk$$

$$\delta(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ixu} dx$$

$$\int_{-\infty}^{\infty} e^{iux} dx = 2\pi\delta(u)$$

Using fourier transform:

$$g(x) = e^{i\omega x}$$

$$\hat{g}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x) e^{-ikx} dx$$

$$\hat{g}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\omega x} e^{-ikx} dx$$

$$\hat{g}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ix(\omega-k)} dx$$

Using what we found earlier

$$\int_{-\infty}^{\infty} e^{ixu} dx = 2\pi\delta(u)$$

$$u = \omega - k$$

$$\int_{-\infty}^{\infty} e^{ix(\omega-k)} dx = 2\pi\delta(\omega - k)$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x) e^{-ikx} dx = \frac{1}{\sqrt{2\pi}} 2\pi\delta(\omega - k)$$

$$\hat{g}(k) = \frac{1}{\sqrt{2\pi}} 2\pi\delta(\omega - k)$$

(c) Use part (b) to find the Fourier transform of $\sin(\omega x)$.

$$\begin{aligned}\sin(x) &= \frac{e^{i\omega x} - e^{-i\omega x}}{2i} \\ \mathcal{F}(\sin(x)) &= \mathcal{F}\left(\frac{e^{i\omega x} - e^{-i\omega x}}{2i}\right) \\ &= \frac{1}{2i}\mathcal{F}(e^{i\omega x}) - \frac{1}{2i}\mathcal{F}(e^{-i\omega x}) \\ &= \frac{1}{2i} \frac{1}{\sqrt{2\pi}} 2\pi\delta(\omega - k) - \frac{1}{2i} \frac{1}{\sqrt{2\pi}} 2\pi\delta(\omega + k)\end{aligned}$$